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DC Motor Torque Loss Diagnostic Checklist

Field procedure for maintenance engineers: find out why a previously adequate PMDC motor now bogs down under load

Work through the steps in order; each one rules out a group of causes before you strip anything down. Record every reading against the motor serial number. Isolate and lock out the supply before uncoupling the motor, handling brushes or blocking the rotor.

Step 1 — Define the complaint

- [ ] Note exactly when torque is lost: at breakaway, on peaks, after warm-up, late in the shift or all the time.
- [ ] Confirm the load has not changed (heavier product, added attachment, steeper ramp). If it has, the drive needs re-sizing, not repair.
- [ ] Collect the nameplate (V, A, rpm, W, duty) and the original test data: no-load speed, no-load current, resistance.
- [ ] Record recent events: stalls, jams, reversing at full speed, overheating, battery or controller replacement.

Step 2 — Supply and circuit (motor loaded)

- [ ] Measure voltage at the motor terminals and at the supply at the same time while the motor carries its normal load.
- [ ] Circuit drop above about 3 % of nominal voltage at rated current needs correcting.
- [ ] Measure the drop across each joint, fuse and contactor pole; a joint dropping several times more than similar joints is faulty.
- [ ] Battery systems: check state of charge and compare each battery's voltage under load; one low block indicates high internal resistance.

Step 3 — Controller current limit

- [ ] Log armature current with a true-DC clamp meter while the machine bogs down.
- [ ] Current rising to a flat plateau while speed collapses means the controller is limiting.
- [ ] Compare the plateau with the controller parameter setting and with the current the load needs: (load torque + friction torque) ÷ Kt.
- [ ] Check for thermal foldback (controller heatsink temperature) and low-voltage current reduction.
- [ ] Never raise the limit above the motor's or gearbox's rated peak.

Step 4 — Cold, decoupled no-load test (magnet health)

- [ ] Let the motor cool to ambient; magnet flux falls about 0.18–0.2 % per °C, so a warm test gives a false result.
- [ ] Uncouple the motor, apply a measured voltage near rated and record terminal voltage, no-load current and speed.
- [ ] Calculate Ke = (V − I0 × R − brush drop) ÷ ω0, with ω0 = 2π × N0 ÷ 60.
- [ ] Ke more than about 5 % below the original value, with higher no-load speed, indicates irreversible flux loss.
- [ ] Check friction torque Ke × I0: unchanged means the motor's mechanics are healthy; higher means bearings or brush friction.

Step 5 — Locked-rotor resistance and brush gear

- [ ] Block the rotor, apply a reduced voltage giving about rated current for no more than a few seconds, and record V and I.
- [ ] Repeat at three or four shaft positions; R = V ÷ I should be consistent and close to the original value.
- [ ] High resistance: inspect brush length against the wear limit, spring force, brush face condition and commutator film.
- [ ] Replace brushes only with the specified grade and bed them to the commutator before applying full load.

Step 6 — Driveline and brake

- [ ] Turn the gearbox output by hand with the motor removed; feel for tight spots, roughness or excessive backlash.
- [ ] Check gearbox oil or grease level and condition; worm gearboxes lose efficiency quickly with worn wheels or degraded lubricant.
- [ ] Measure brake coil voltage while released and check the air gap against the brake manufacturer's setting.
- [ ] If the decoupled motor tests normal, the torque loss is downstream: repair the driveline before replacing the motor.

Ke back-calculation worksheet (worked example, 24 V PMDC motor)

Quantity	Formula	Original test data	Today (cold, decoupled)	Result
Terminal voltage V	Measured	24.0 V	24.0 V	Same test voltage
No-load current I0	Measured	1.20 A	1.35 A	+12.5 %
No-load speed N0	Measured	3,350 rpm	3,750 rpm	+11.9 %
Angular speed ω0	2π × N0 ÷ 60	350.8 rad/s	392.7 rad/s	—
Back-EMF V − I0R − Vb	R = 0.18 Ω, Vb = 1.0 V	22.78 V	22.76 V	—
Ke = Kt	Back-EMF ÷ ω0	0.0650 V·s/rad	0.0580 V·s/rad	−10.8 %: flux loss
Friction torque	Ke × I0	0.078 Nm	0.078 Nm	Unchanged: mechanics OK

Values are typical guidance for fractional-kW PMDC motors. Always compare against the motor's own nameplate and original test data, and keep locked-rotor tests short to avoid heating the winding or stressing the magnets.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.